

ICOSAGEN RESEARCH ARTICLE

Broadly Neutralizing Humanized Antibody for SARS- CoV-2

Effective Pan-Neutralization and Enhanced Protection through Inhalation-Based
Delivery in Non-Human Primates

Introduction & The Challenge

| Limitations of Current Therapeutics

Systemic Delivery Barriers

Standard intravenous (IV) administration is cumbersome and costly. Achieving protective concentrations in mucosal epithelia—the primary site of infection—requires extremely high systemic doses to facilitate FcRn-mediated transport.

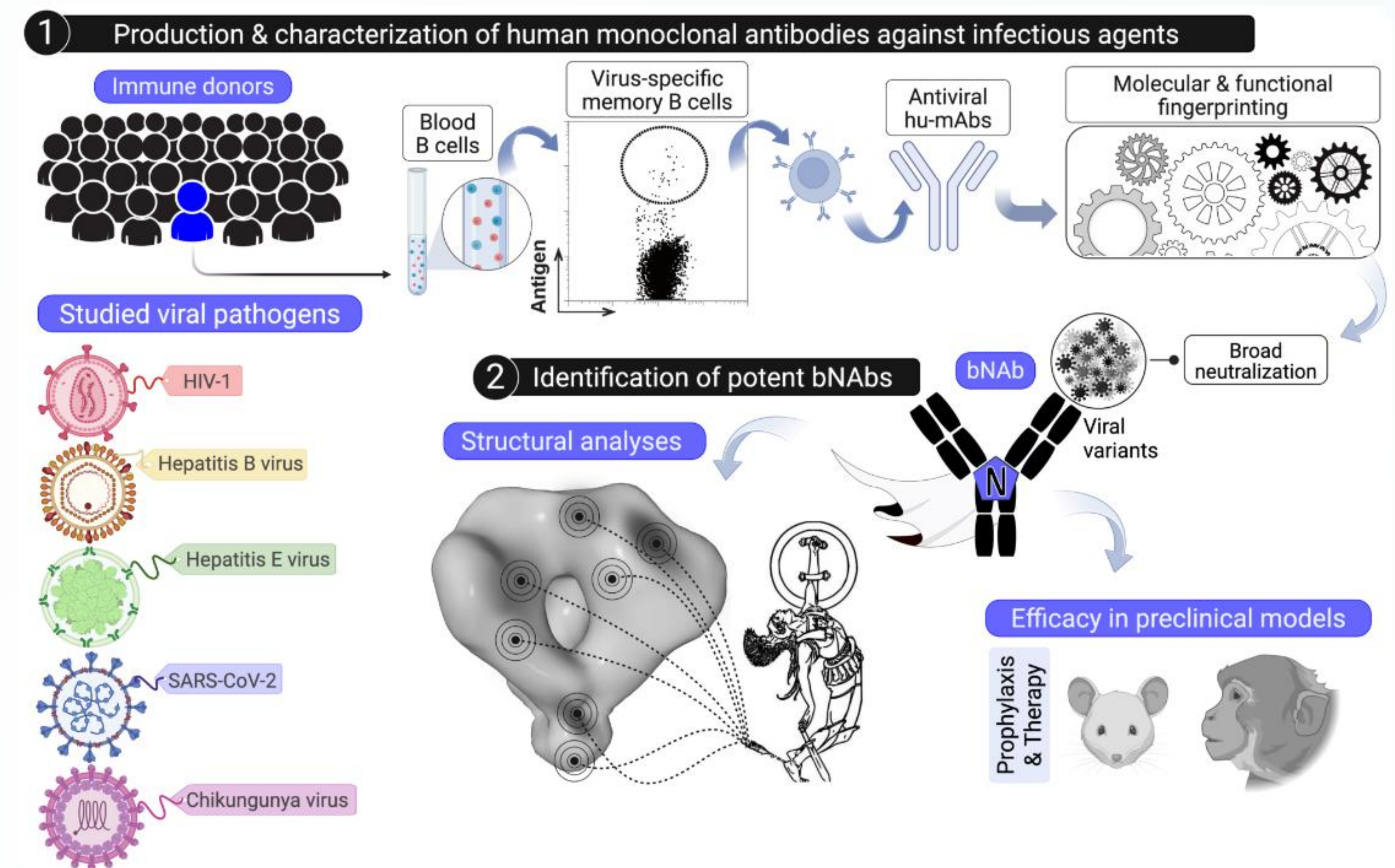
Viral Mutational Escape

SARS-CoV-2 variants of concern (VoCs) exhibit significant mutational plasticity. Most first-generation monoclonal antibodies lose efficacy against emerging strains like Omicron due to mutations in the Spike protein RBD.

Antibody Discovery via HybriFree

The HybriFree platform enables the rapid isolation of high-affinity antibodies without the need for B-cell culturing or single-cell manipulation.

- **ICO-hu104:** Derived from immunized rabbits and humanized via the Qumanize algorithm.
- **Developability:** Maintains a humanization score > 85% with optimized biophysical properties.
- **Precision:** Screened against Beta (B.1.351) to ensure resistance to common mutational hot-spots.



| Unmatched Neutralizing Potency

100%

Pan-VoC Neutralization

Sub-Nanomolar IC₅₀

ICO-hu104 preserves high neutralizing potency across all major variants of concern. Pseudovirus assays confirm picomolar to low-nanomolar efficacy against Wuhan, Alpha, Beta, Delta, and multiple Omicron sub-lineages (BA.2, BQ.1, BQ.1.1).

Delivery & Efficacy Study

| The Inhalation Delivery Approach

Nebulization (Neb.) offers a non-invasive alternative for direct delivery to the respiratory tract.

- **Device:** Aerogen Solo vibrating mesh nebulizer retains structural integrity and functionality of the antibody.
- **Targeting:** Immediate high drug concentration at the primary site of viral replication.
- **Accessibility:** Lower overall dose required, reduced costs, and minimal systemic side effects.



Distribution in Lung Tissue: IV vs Nebulization

IV Concentration (BAL)



Nebulization (BAL)



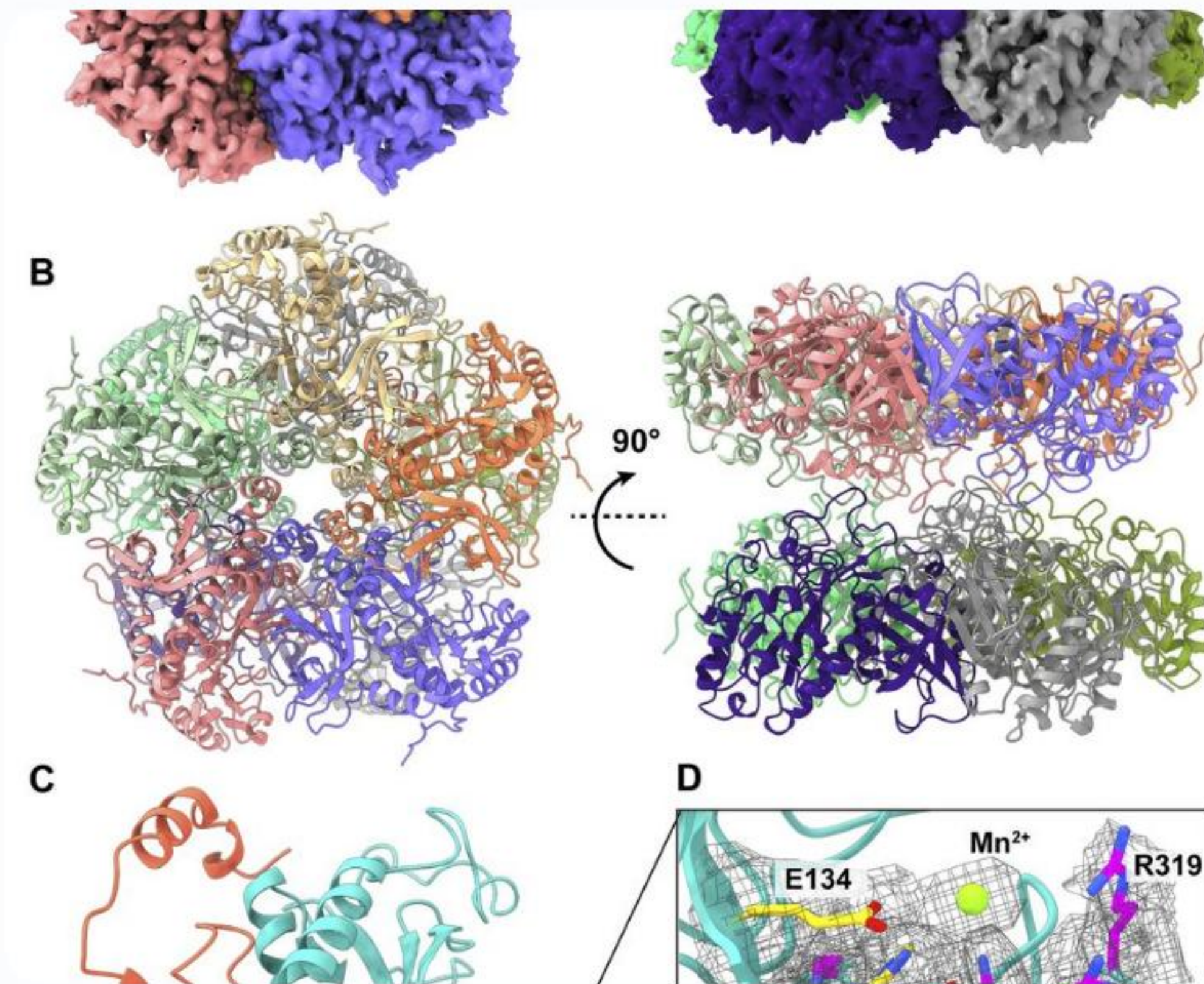
Result: Nebulization yields a nearly 100-fold increase in protective antibody levels within Bronchoalveolar Lavage (BAL) compared to systemic delivery.

Antiviral Protection in NHPs

In a non-human primate (macaque) study, a single dose of 25 mg/kg nebulized ICO-hu104 significantly inhibited SARS-CoV-2 Delta replication.

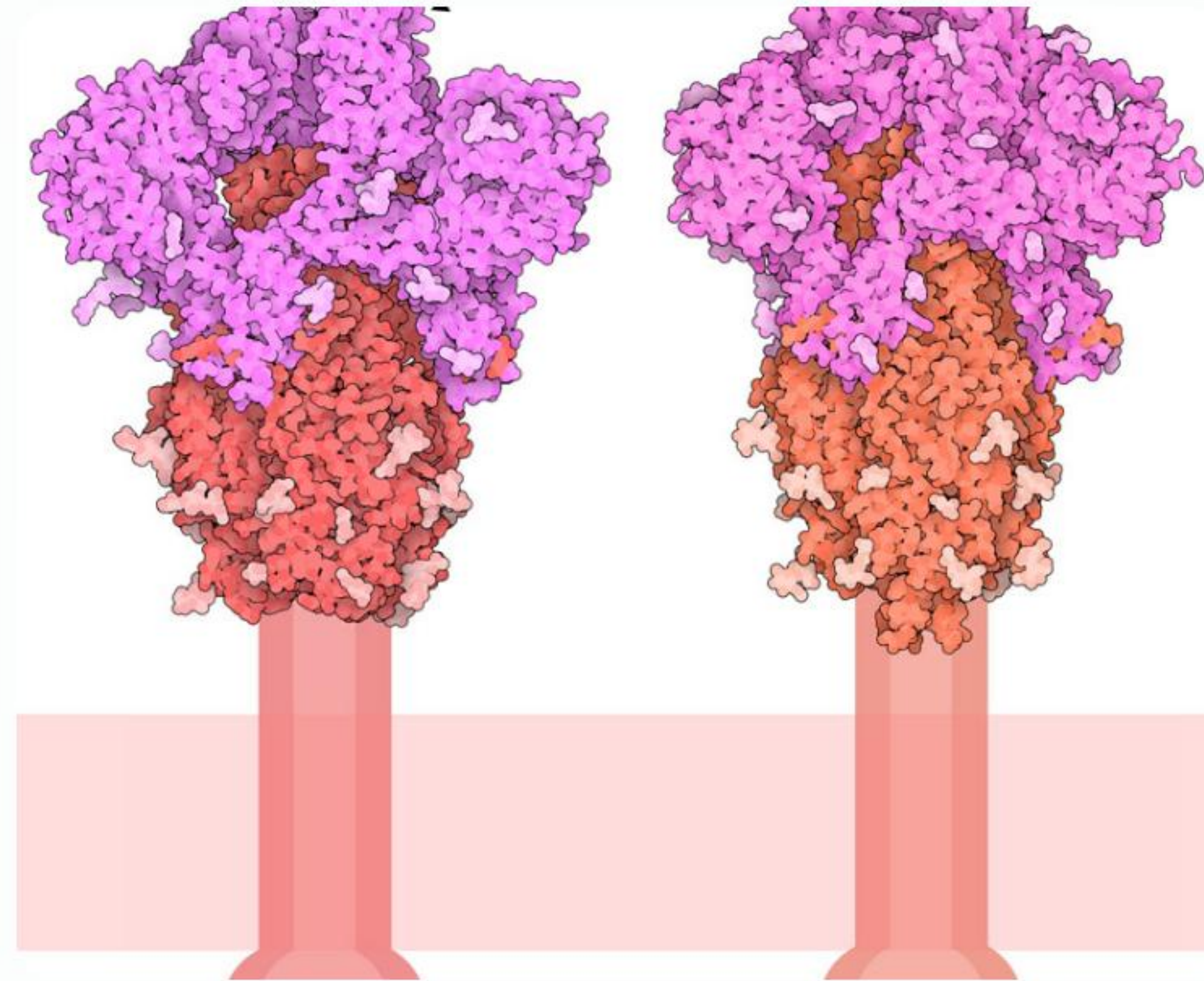
- Significant reduction in genomic RNA (gRNA) in nasopharyngeal and tracheal fluids.
- Marked decrease in subgenomic RNA (sgRNA), indicating inhibited viral replication.
- Higher mucosal concentration directly correlated with lower viral load AUC.

Structural & Functional Verification



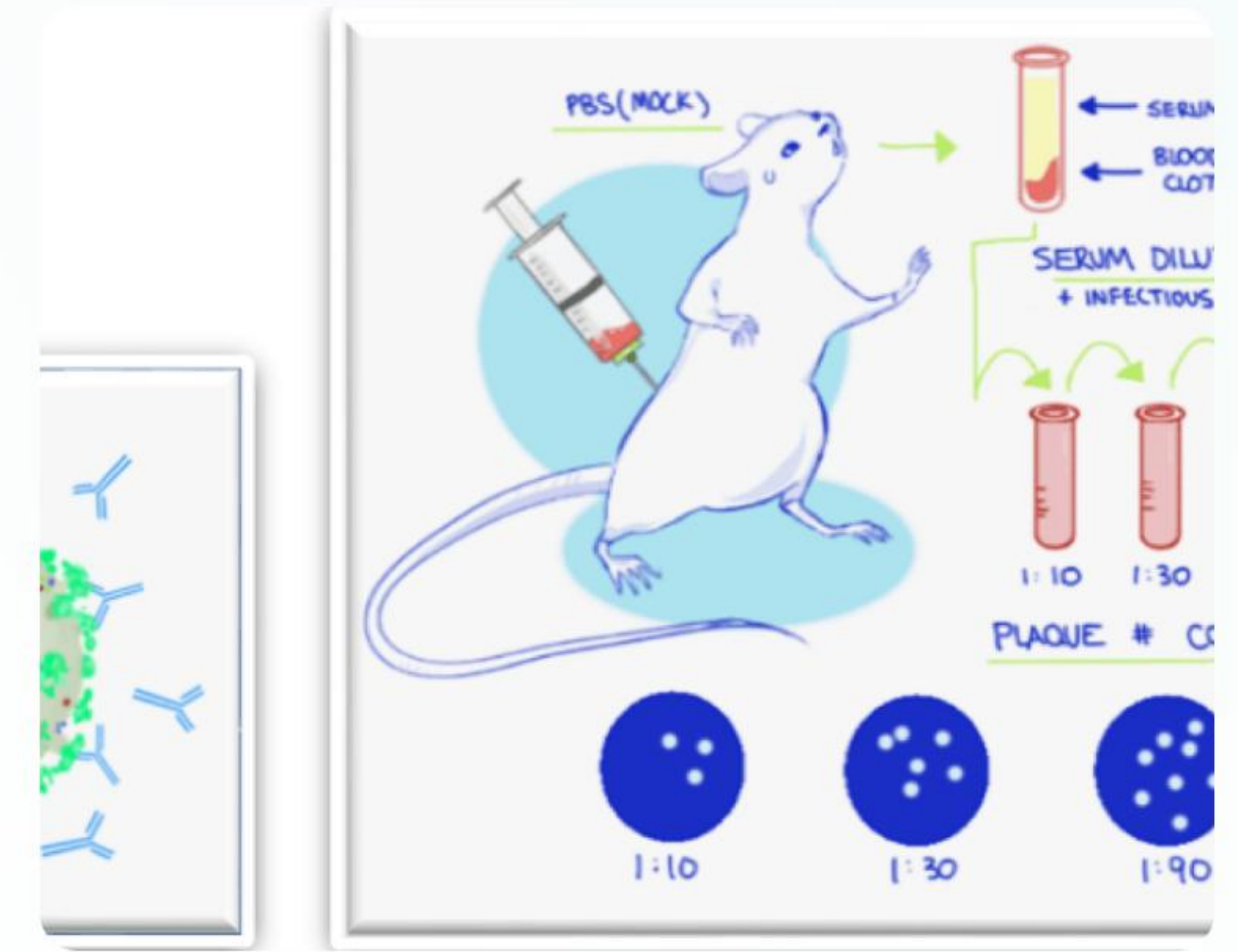
Cryo-EM Analysis

Complexes reveal Fabs binding to RBD in the "down" conformation.



Spike Epitope

Targeting a conserved beta-sheet region removed from common mutation sites.



Functional Integrity

Antibody remains 100% active post-nebulization shear forces.

Epitope Conservation: ICO-hu104 vs Predecessors

Feature	ICO-hu23 (1st Gen)	ICO-hu104 (Next Gen)
Primary Source	Convalescent Human Serum	Immunized Rabbit (Humanized)
Epitope Region	Loop 472-502 (RBD Tip)	Beta Sheet 433-443 (RBD Side)
Omicron Status	Lost Binding (6 Mutations)	Maintained (Zero Mutations)
Neutralization Type	Orthosteric (ACE2 Blocking)	Allosteric / Non-RBD Interface

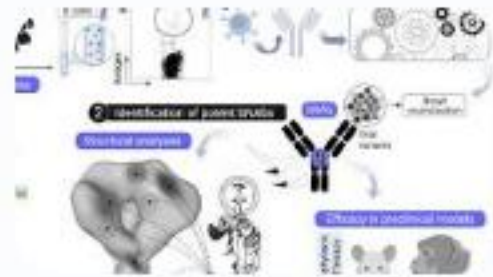
Questions?

Thank you for your attention.

Contact: mart.ustav.jr@icosagen.com

Icosagen Cell Factory | Tartu, Estonia

Image Sources



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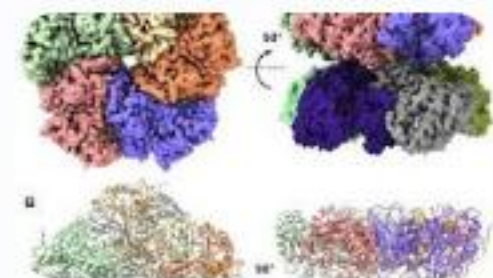
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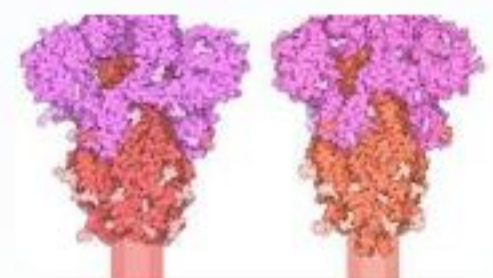
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